CPTG and the Cosmological Horizon Problem

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Abstract

The cosmological horizon problem arises because widely separated regions of the cosmic microwave background exhibit near-uniform temperature even though, in a non-inflationary Friedmann expansion, they appear unable to have exchanged ordinary causal signals before decoupling. Standard cosmology resolves this by introducing an early inflationary epoch in which a small causally connected region expands rapidly into the observable universe. Curvature Polarization Transport Gravity, or CPTG, provides a different geometric route. In CPTG, the relevant early-universe scale is not a scalar inflaton field and not the raw native pi-branch Hubble normalization. It is the structural Hubble-equivalent scale associated with the time-dependent curvature-response scale $a_\star(t)$. On this branch, horizon-scale uniformity is treated as curvature-state synchronization: finite-density curvature saturation prevents divergent early collapse, while active geometric transport organizes the metric response before matter decoupling. The mechanism therefore shifts the horizon problem from ordinary thermal contact between distant matter regions to synchronization of the primordial curvature state. This produces falsifiable consequences in CMB anisotropy statistics, primordial non-Gaussianity, tensor backgrounds, and the relation between structural acceleration response and cosmological expansion.

1 The Horizon Problem

The cosmic microwave background is nearly uniform across the sky. Its mean temperature is approximately the same in directions separated by angles so large that, under an ordinary non-inflationary Friedmann expansion history, those regions appear not to have been in causal contact before recombination. The small anisotropies observed in the microwave background are crucial for structure formation, but their amplitude is tiny compared with the uniform background temperature [1, 2, 3].

The standard resolution is cosmic inflation. A small region that was initially in causal contact undergoes accelerated expansion, stretching to a size larger than the observable universe. After inflation ends, the hot big-bang phase begins with nearly homogeneous initial conditions already in place [4, 5, 6]. Inflation also provides a mechanism for generating nearly scale-invariant primordial perturbations, which are later observed through the cosmic microwave background power spectrum.

CPTG [10, 11, 12] approaches the same problem from a different starting point. The early universe is not treated as an initially singular point requiring a separate inflation stage to smooth causal patches. Instead, the primordial state is modeled as a finite-density, gradient-saturated geometric boundary. Homogeneity is produced by the coupled action of curvature polarization and directional transport before matter decoupling.

This paper uses the CPTG-native ordering: nonlinear curvature structure first, structural reduction second, and observational comparison coordinates third. The horizon mechanism is therefore not built from an independently fitted observational Hubble constant. It is built from the structural curvature-rate scale associated with the epoch-dependent acceleration response. The raw native pi-branch normalization, the transported CMB comparison value, and the structural Hubble-equivalent scale play different roles and are kept separate below.

2 Parameter and Symbol Guide

The horizon mechanism uses several Hubble-like and acceleration-like quantities that belong to different CPTG layers. Tables 1, 2, and 3 fix the notation used in this paper so that the structural horizon argument is not confused with the pi-branch comparison coordinates.

Symbol	Meaning in this paper
$a_\star(t)$	Epoch-dependent structural acceleration scale controlling primordial curvature response. It is not automatically identical to the fixed pi-branch anchor.
$a_{\star,0}$	Present-epoch value of the structural scale used only to calibrate the structural branch relation.
$H_{\rm struct}(t)$	Curvature-rate scale locked to $a_\star(t)$ through $H_{\rm struct}(t) = (16\pi/3)a_\star(t)/c$.
$H_{\rm struct,0}$	Present-epoch structural Hubble-equivalent value. It should not be collapsed into the raw pi-branch or CMB comparison value.
$\eta_H(t)$	Dimensionless structural locking ratio $cH_{\rm struct}(t)/a_\star(t) = 16\pi/3$ for the horizon branch.
$\ell_\star(t)$	Local curvature-response length $c^2/a_\star(t)$.
$\tau_\star(t)$	Local structural time $c/a_\star(t)$.
C	Reduced curvature-response invariant containing field strength and spatial-gradient organization.
χ	Dimensionless response $C/a_\star^2(t)$ controlling saturation.
$\chi_{\rm grad}$	Gradient-dominated part of χ used to describe finite primordial saturation.
S^α	Directional curvature-transport vector controlling how organized curvature response propagates.
Σ	Coherence or transport-state variable describing organized curvature synchronization.
Φ	Scalar potential used in the reduced transport expression.
$a_{\rm min}$	Nonzero minimum scale factor in the finite-boundary representation.
A_1, A_2	Boundary expansion coefficients distinguishing symmetric bounce behavior from one-sided emergence near the saturated boundary.

Table 1: Structural horizon-branch symbols used in the CPTG horizon mechanism.

Symbol	Meaning in this paper
$a_{\pi,0}$	Fixed geometric acceleration anchor of the cosmic pi branch. It defines the native branch normalization, not the horizon structural scale by itself.
H_∞	Asymptotic pi-branch scale $a_{\pi,0}/c$.
$H_0^{(\pi)}$	Raw present-epoch geometric normalization of the pi branch.

Table 2: Native pi-branch symbols kept distinct from the structural horizon branch.

Symbol	Meaning in this paper
$H_{0,\mathrm{CMB}}^{\mathrm{CPTG}}$	Transported acoustic/CMB Hubble comparison value. It is an observational-coordinate projection, not the structural horizon scale.

Table 3: Transported CMB comparison-coordinate symbol used to separate observational projection from the structural horizon scale.

3 Structural Acceleration Scaling

The first CPTG ingredient is not the present-day Hubble constant by itself, but the theory's structural acceleration scale. This scale governs how curvature response organizes across local weak-field systems and across the cosmological background. In the core CPTG hierarchy, the local structural scale $a_\star$ is not identical to the fixed cosmic pi-branch anchor $a_{\pi,0}$; it is the acceleration scale associated with the structure or branch being modeled. In the early-universe application, the structural scale is allowed to be epoch dependent:

$$a_\star = a_\star(t). \tag{1}$$

The associated structural length is

$$\ell_\star(t) = \frac{c^2}{a_\star(t)}. \tag{2}$$

The associated structural time is

$$\tau_\star(t) = \frac{c}{a_\star(t)}. \tag{3}$$

The structural Hubble-equivalent branch is then written as an epoch-dependent CPTG scaling relation:

$$H_{\mathrm{struct}}(t) = \frac{16\pi}{3}\frac{a_\star(t)}{c}. \tag{4}$$

Equivalently,

$$H_{\mathrm{struct}}(t) = \frac{16\pi}{3}\frac{1}{\tau_\star(t)}. \tag{5}$$

The present-day structural Hubble-equivalent relation is recovered only as the late-time calibration of this scaling branch:

$$H_{\mathrm{struct}}(t_0) = H_{\mathrm{struct},0}. \tag{6}$$

At the present epoch,

$$a_\star(t_0) = a_{\star,0}. \tag{7}$$

Therefore,

$$H_{\rm struct,0} = \frac{16\pi}{3}\frac{a_{\star,0}}{c}. \tag{8}$$

This structural scale should not be collapsed into the separate pi-branch comparison quantities. The native cosmic branch has its own fixed geometric anchor $a_{\pi,0}$, asymptotic scale $H_\infty = a_{\pi,0}/c$, raw present-epoch normalization $H_0^{(\pi)}$, and transported acoustic comparison value $H_{0,\rm CMB}^{\rm CPTG}$. The quantity $H_{\rm struct}(t)$ used here is the structural curvature-rate scale tied to $a_\star(t)$.

This ordering keeps the early-universe argument tied to CPTG's structural scale rather than to an observational constant measured at the present epoch. The same structural scale that controls weak-field curvature response also controls the cosmological branch, but the horizon problem is governed by the primordial value of the scale rather than by inserting today's Hubble constant into the early universe.

The dimensionless form of the scaling relation is

$$\eta_H(t) = \frac{cH_{\rm struct}(t)}{a_\star(t)}. \tag{9}$$

For the CPTG scaling branch,

$$\eta_H(t) = \frac{16\pi}{3}. \tag{10}$$

This fixed ratio is a structural boundary condition for the cosmological sector. It states that the expansion scale and the acceleration-response scale remain locked together by geometry across the branch, while their absolute values may evolve with epoch.

4 Curvature Saturation at the Primordial Boundary

The second ingredient is the CPTG curvature invariant. In the cosmological application, this invariant is used as the quasi-static or perturbative curvature-response limit of the full covariant structure. It contains both field strength and spatial field variation:

$$C = g_j g_j + \kappa^2(\partial_i g_j)(\partial_i g_j). \tag{11}$$

The invariant is normalized by the epoch-dependent acceleration scale:

$$\chi(t) = \frac{C}{a_\star^2(t)}. \tag{12}$$

The gradient component is

$$\chi_{\text{grad}}(t) = \frac{\kappa^2(\partial_i g_j)(\partial_i g_j)}{a_\star^2(t)}. \tag{13}$$

The polarization sector is written as

$$\mathcal{L}_{\text{pol}} = \beta a_\star^2(t)\chi^{1/3}. \tag{14}$$

The derivative with respect to the spatial field gradient is

$$\frac{\partial \mathcal{L}_{\text{pol}}}{\partial(\partial_i g_j)} = \frac{2}{3}\beta\chi^{-2/3}\kappa^2(\partial_i g_j). \tag{15}$$

When the gradient term dominates,

$$\chi \approx \chi_{\text{grad}}. \tag{16}$$

The high-gradient response then softens:

$$\lim_{\partial_i g_j \to \infty} \frac{\partial \mathcal{L}_{\text{pol}}}{\partial(\partial_i g_j)} = 0. \tag{17}$$

This behavior replaces the singular backward extrapolation with a saturated geometric boundary. The early universe does not collapse to a zero-volume point in the CPTG description. It approaches a finite, highly warped state in which gradient growth is dynamically bounded.

A regularized primordial scale factor can be represented schematically by

$$a(\tau) = a_{\min} + A_1\tau + \frac{1}{2}A_2\tau^2 + \mathcal{O}(\tau^3). \tag{18}$$

The corresponding minimum condition is

$$a_{\min} > 0. \tag{19}$$

A symmetric bounce branch is represented by

$$A_1 = 0. \tag{20}$$

A one-sided emergence branch is represented by

$$A_1 > 0. \tag{21}$$

This expansion is not introduced as an inflation-driven exponential phase. It is the emergence of expansion from a finite curvature-saturated boundary.

5 Directional Curvature Transport

The third ingredient is the transport sector. CPTG introduces a directional vector field that governs how organized curvature structure is redistributed across the geometry. The transport Lagrangian is

$$\mathcal{L}_{\text{trans}} = -\frac{\lambda_t}{2} \left(S^\alpha \Sigma \nabla_\alpha \Phi \right)^2 . \tag{22}$$

The transport vector is normalized as

$$S^\alpha S_\alpha = \pm 1. \tag{23}$$

The suppression factor is bounded by

$$0 \le \Sigma \le 1. \tag{24}$$

The active transport branch is represented by

$$T_+ : \quad \Sigma \to 1. \tag{25}$$

The saturated branch is represented by

$$T_0 : \quad \Sigma \to 0. \tag{26}$$

In the primordial high-density state, matter remains locally constrained by the speed of light. The transport sector does not require ordinary particles to move faster than light. Instead, it describes the redistribution of the geometric state itself. The field variable being smoothed is the curvature organization of the metric, not a material signal traveling through space after recombination.

The transport timescale is expressed through a geometric relaxation rate rather than through an ordinary signal velocity:

$$\tau_{\text{trans}}(L, t) \sim \Gamma^{-1}_{\text{trans}}(L, \chi, \Sigma, S^\alpha). \tag{27}$$

The ordinary light-crossing time is

$$\tau_c = \frac{L}{c}. \tag{28}$$

The horizon problem is avoided when geometric transport acts across the relevant CMB smoothing scale before matter decoupling:

$$\tau_{\text{trans}}(L_{\text{CMB}}, t_{\text{dec}}) < \tau_{\text{dec}}. \tag{29}$$

This is not an inflationary condition. It is a curvature-synchronization condition.

6 Causal Contact Without Inflation

The standard horizon problem assumes that thermal equilibrium requires ordinary causal exchange between separated matter regions. CPTG replaces that assumption with a two-layer causal structure. Matter remains locally causal, while geometry undergoes active transport during the primordial boundary phase.

The relevant smoothing variable is the curvature-response field:

$$\mathcal{G}(x,t) = (\chi,\, \chi_{\mathrm{grad}},\, S^{\alpha},\, \Sigma,\, a_{\star}(t)) \,. \tag{30}$$

Primordial homogeneity corresponds to relaxation of this field toward a common geometric state:

$$\mathcal{G}(x,t) \to \overline{\mathcal{G}}(t). \tag{31}$$

Temperature uniformity then follows because matter evolves inside an already synchronized geometric background. The microwave background is uniform not because every later-separated matter region exchanged photons across the entire observable universe, but because the curvature state setting the thermal background had already been flattened by transport before decoupling.

The CPTG horizon condition can be summarized as

$$L_{\mathrm{smooth}}(t_{\mathrm{dec}}) \geq L_{\mathrm{CMB}}(t_{\mathrm{dec}}). \tag{32}$$

Here, the smoothing length is determined by curvature transport rather than by ordinary photon propagation alone.

7 Relation to Standard Inflation

Inflation solves the horizon problem by expanding a small, causally connected patch into a much larger region. Its most studied implementations also predict a nearly scale-invariant scalar spectrum and a tensor background whose amplitude is commonly expressed through the tensor-to-scalar ratio [7, 9].

CPTG does not reproduce inflation by adding another scalar field. It changes the pre-recombination causal mechanism. The comparison can be written as

$$\begin{array}{cc}
\textbf{Inflation} & \textbf{CPTG} \\
\text{causal patch} & \text{finite curvature boundary} \\
\downarrow & \downarrow \\
\text{exponential expansion} & \text{active geometric transport} \\
\downarrow & \downarrow \\
\text{homogeneous CMB} & \text{homogeneous CMB}
\end{array} \tag{33}$$

Both frameworks must match the same cosmic microwave background observations. The distinction lies in the origin of the initial conditions. Inflation smooths by stretching. CPTG smooths by transport and saturation.

8 Perturbation Structure

The strongest observational distinction appears in the perturbation sector. Inflationary models typically produce nearly Gaussian, nearly adiabatic, nearly scale-invariant scalar perturbations, with stringent observational constraints from Planck [7, 8]. CPTG produces perturbations from nonlinear curvature relaxation and directional transport.

The primordial curvature perturbation can be written schematically as

$$\mathcal{R}_{\text{CPTG}} = \mathcal{R}_{\text{sat}} + \mathcal{R}_{\text{trans}}. \tag{34}$$

The saturation contribution follows from the bounded curvature invariant:

$$\mathcal{R}_{\text{sat}} = F(\chi, \chi_{\text{grad}}). \tag{35}$$

The transport contribution follows from the directional sector:

$$\mathcal{R}_{\text{trans}} = G(S^{\alpha}, \Sigma, \nabla_{\alpha}\Phi). \tag{36}$$

The scalar power spectrum can then be represented as

$$P_{\mathcal{R}}(k) = P_{\text{sat}}(k) + P_{\text{trans}}(k) + 2P_{\text{cross}}(k). \tag{37}$$

The cross term is important. It allows CPTG to predict correlated structure between curvature saturation and directional transport. This provides a direct route to non-Gaussian or anisotropic signatures.

A schematic bispectrum diagnostic is

$$B_{\mathcal{R}}(k_1, k_2, k_3) = \langle \mathcal{R}_{\text{CPTG}}(k_1)\mathcal{R}_{\text{CPTG}}(k_2)\mathcal{R}_{\text{CPTG}}(k_3) \rangle . \tag{38}$$

Because the source is nonlinear curvature transport, CPTG does not naturally require this bispectrum to match the simplest slow-roll templates. Planck's strong constraints on primordial non-Gaussianity therefore become a direct test of the allowed transport sector [8].

The allowed CPTG bispectrum must remain below observational limits:

$$\left| B_{\mathcal{R}}^{\text{CPTG}} \right| \leq B_{\mathcal{R}}^{\text{obs}}. \tag{39}$$

9 Tensor Background

Inflation commonly predicts a primordial tensor background generated by quantum fluctuations stretched to cosmic scales. Observational searches for large-angular-scale B-mode polarization constrain the allowed tensor-to-scalar ratio [9, 7].

CPTG predicts a different tensor source. Tensor perturbations arise from relaxation of anisotropic curvature transport and from phase transitions in the early geometric state. The tensor spectrum may be written schematically as

$$P_T^{\mathrm{CPTG}}(k) = P_T^{\mathrm{relax}}(k) + P_T^{\mathrm{phase}}(k). \tag{40}$$

The relaxation contribution is tied to the transport vector:

$$P_T^{\mathrm{relax}}(k) = Q(S^\alpha, \Sigma, k). \tag{41}$$

The phase-transition contribution is tied to changes in the active transport branch:

$$P_T^{\mathrm{phase}}(k) = R(T_+ \to T_0, k). \tag{42}$$

The resulting gravitational-wave background does not need to share the same scale dependence as slow-roll inflation. It may peak at different frequencies or carry directional memory from the transport field. This provides a falsifiable distinction: CPTG must match the observed absence of a large primordial B-mode signal while allowing a separate transport-generated tensor spectrum.

At CMB scales, the CPTG tensor contribution must satisfy

$$P_T^{\mathrm{CPTG}}(k_{\mathrm{CMB}}) \leq P_T^{\mathrm{obs}}(k_{\mathrm{CMB}}). \tag{43}$$

A transport-generated tensor component may therefore be hidden below current CMB B-mode limits, shifted toward non-CMB frequency windows, or suppressed by the same transport saturation that smooths the primordial field.

10 Observable Signatures

CPTG makes several linked cosmological predictions.

First, the structural acceleration scale and the structural expansion-rate branch must remain locked:

$$H_{\mathrm{struct}}(t) = \frac{16\pi}{3}\frac{a_\star(t)}{c}. \tag{44}$$

The present-day structural relation is the late-time limit:

$$H_{\mathrm{struct},0} = \frac{16\pi}{3}\frac{a_{\star,0}}{c}. \tag{45}$$

This is a structural curvature-rate relation. It must be compared to observational Hubble determinations only through the appropriate CPTG comparison-coordinate map, not by identifying it directly with every reported Hubble summary.

Second, the microwave background should be homogeneous because the primordial curvature state relaxes before decoupling:

$$\mathcal{G}(x, t_{\text{dec}}) \approx \overline{\mathcal{G}}(t_{\text{dec}}). \tag{46}$$

Third, residual non-Gaussianity should track nonlinear curvature transport rather than a purely scalar inflaton field:

$$B_{\mathcal{R}}^{\text{CPTG}} \neq B_{\mathcal{R}}^{\text{single field}}. \tag{47}$$

Fourth, the tensor background should be tied to transport relaxation rather than only to inflationary vacuum stretching:

$$P_T^{\text{CPTG}}(k) \neq P_T^{\text{slow roll}}(k). \tag{48}$$

Fifth, the primordial boundary should avoid a zero-volume singularity:

$$a_{\min} > 0. \tag{49}$$

These predictions are mutually constrained. The same transport parameters that smooth the horizon must also satisfy microwave-background anisotropy limits, non-Gaussianity limits, tensor-mode constraints, and late-time acceleration-scale tests.

11 Falsifiability

The CPTG horizon solution is falsifiable in several ways.

A measured violation of the late-time structural acceleration-expansion scaling relation would rule out the proposed present-epoch structural branch:

$$\frac{cH_{\text{struct},0}}{a_{\star,0}} \neq \frac{16\pi}{3}. \tag{50}$$

A failure of the epoch-dependent structural scaling law would rule out the corresponding horizon branch:

$$\frac{cH_{\text{struct}}(t)}{a_\star(t)} \neq \frac{16\pi}{3}. \tag{51}$$

A separate comparison-coordinate failure would occur if the native pi-branch quantities, after their locked transport maps, could not reproduce the CMB acoustic, clock, or distance observables required by the larger CPTG branch. This distinction matters: the structural horizon scale, the raw native pi-branch normalization, and the transported CMB comparison normalization are related CPTG layers, not interchangeable symbols.

A microwave-background bispectrum fully consistent with only the simplest slow-roll scalar templates and excluding all transport-correlated shapes would strongly restrict the CPTG transport sector. A tensor spectrum matching the slow-roll inflationary form with no room for transport relaxation would similarly restrict the model. Conversely, a statistically significant non-Gaussian or tensor residual correlated with a transport-like anisotropic structure would favor the CPTG mechanism.

The theory is also constrained by the absence of excessive CMB anisotropy. Transport must smooth the primordial field without leaving too much directional memory. This requires

$$\delta\mathcal{G}_{\text{residual}} \ll \overline{\mathcal{G}}. \tag{52}$$

The horizon solution therefore cannot be tuned independently from the perturbation sector. The same mechanism that solves the horizon problem must also produce the observed small amplitude of cosmic microwave background fluctuations.

12 Conclusion

CPTG resolves the causal form of the horizon problem by replacing inflationary stretching with curvature saturation and active geometric transport. The early universe begins not as a zero-volume singularity but as a finite, gradient-bounded primordial state. During this state, curvature polarization prevents divergent collapse while the transport sector redistributes geometric structure before matter decoupling. Matter remains locally causal, but the curvature-response state of the metric is synchronized by the theory's transport dynamics.

The primitive early-universe quantity is the CPTG structural scale, not a present-day observational Hubble constant. The relation between structural acceleration response and a structural Hubble-equivalent scale appears as the late-time calibration of a deeper epoch-dependent branch. The raw native pi-branch normalization and the transported CMB comparison normalization remain separate comparison layers. This makes the horizon problem a question of structural curvature response rather than a direct insertion of present-day expansion into the primordial universe.

The result is a non-inflationary path to a homogeneous cosmic microwave background. The price of that alternative is testability. CPTG must reproduce the observed microwave-background power spectrum, satisfy non-Gaussianity bounds, remain compatible with tensor-mode limits, and preserve the appropriate structural and comparison-coordinate relations between acceleration response and cosmological expansion. If those constraints are met simultaneously, the horizon problem becomes a geometric transport problem rather than an inflation problem.

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